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Smart CCTV Exam Monitoring to Detect and Alert Suspicious Activities Using Deep Transfer Learning

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Abstract

Examinations are essential in assessing learning outcomes and evaluating students' academic performance. Academic dishonesty, including cheating during exams, violates academic integrity and is a critical issue in educational settings. This study presents an approach to detecting exam cheating by developing an intelligent CCTV system for exam monitoring, capable of issuing alerts for suspicious actions indicative of cheating. The system can detect cheating behaviors, including referencing books/notes/papers, talking to others in the room, accessing the internet, seeking answers from friends via phone calls, and using mobile phones or other electronic devices. The dataset used in this experiment is imbalanced and consists of multiple classes. Deep transfer learning is applied as an advanced feature extraction technique, leveraging prior learning rather than starting from scratch. This study integrates ResNet50V2 with three deep learning architectures, Gate Recurrent Unit (GRU), Long Short-Term Memory (LSTM), and Convolutional Neural Network (CNN). These integrated models were then compared with other transfer learning models, including VGG16, VGG19, ResNet50V2, Inception, and EfficientNetB7. The experimental results reveal that the ResNet50V2 setup yields the best overall performance, with the GRU and LSTM variants attaining the highest F1 scores and accuracies. Both scores and accuracies of ResNet50V2-GRU and ResNet50V2-LSTM models were exactly 1.0 contrasted with the CNN variant which obtained about 0.4151 for F1 score and 0.7232 for accuracy. Compared to other models such as VGG16-LSTM, VGG16-GRU, and VGG16-CNN, the F1 scores achieved only in that order as low as 0.141, 0.3255, and 0.141. Also, the accuracies of VGG16 models are 0.5446, 0.7679, and 0.5446. The examination of model training and testing duration indicates that ResNet50V2-GRU model performs faster than ResNet50V2-LSTM model. Overall, the ResNet50V2-GRU model exceed all the tested model in the paper in term of the classification effectiveness, model training time, and model testing time. Additionally, the study found that deep transfer learning can effectively handle imbalanced datasets without requiring data resampling during preprocessing.

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1. Introduction

Cheating in exams is a problem in education that occurs in all countries and levels. Exams are an assessment process of the learning process, so they are important for evaluating learning outcomes. Cheating during exams is one of the violations of academic integrity. To minimize this, the government and education experts innovate in the learning model where assessments are carried out not by exams but by project-based, case studies, or administrative matters. However, it is necessary to realize that not all subjects can be applied to project-based or case studies. The selection process for university entrances and employees still involves exams and tests, both written and computer-based [1].

Proctoring during exams is crucial to prevent cheating. The absence or negligence of the on-site exam proctor may lead to dishonest behavior of the test takers. As a human, a proctor can interpret the examinee's movements as suspicious actions by observing their gestures. However, full supervision during exams by human proctors is highly time-consuming, particularly for teachers and lecturers, which often leads to decreased academic productivity. Additionally, humans have limited fields of vision and restricted movement, which can hinder their ability to provide continuous supervision. The proctor's movement around the exam room can create loopholes for test takers to cheat. Seeing this phenomenon, technological innovations, artificial intelligence, and computer vision are urgently needed to aid exam proctoring and provide cheating alerts [2],[3].

Detecting cheating during exams is challenging because identifying test takers' activities from video footage is complicated. The video data must first be converted into multiple digital image frames. From this collection of frames, a subset of informative frames or key frames is then extracted for further analysis. However, keyframe selection remains an unresolved issue. Mingju Chen and his colleagues applied object detection and image quality techniques to select key frames from video footage. The application of deep learning model for classification has demonstrated outstanding results, as supported by findings from several other studies [4], [5], [6].

Nischal et al. developed an automated exam monitoring and fraud detection system to address the limitations of manual cheating detection methods. Their proposed system works by analyzing students' postures and emotions during exams using computer vision technology. It utilizes surveillance cameras equipped with thermal sensors and eye-tracking devices to monitor behavior in real time [7]. Cheating detection based on video recording has evolved significantly with the integration of advanced technologies, particularly deep learning and multimodal analysis. These approaches enhance the accuracy and efficiency of detecting dishonest behaviors during online examinations. Utilize deep convolutional neural networks (CNNs) to analyze video and audio frames, for example, can effectively identify cheating behavior [8]. Integrated multiple modules, including video analysis of front and rear cameras and speech recognition, enables comprehensive monitoring throughout the exam.

Automated video proctoring reduces the need for human oversight, making the process less labor-intensive and more scalable [9]. In the preliminary study on digital exam cheating detection, the automated detection system was able to maintain high accuracy in identifying cheating, outperforming the manual process of having a human review each examinee's screen recording individually [10]. Multimodal Fusion Techniques, which combine video, audio, and text data, improve deception detection rates by up to 13% compared to relying on a single data source application [11]. Additionally, micro expression analysis can greatly enhance the detection of deceptive behavior.

While these advancements present a robust framework for cheating detection, concerns about privacy and the potential for false positives remain critical issues that need addressing in future implementations. A study by Alrubaish and team proposes modification of existing algorithms to improve the model's ability to detect students' cheating intent during exams. The study used a surveillance camera equipped with a thermal detector and an eye tracking system. When a test taker attempts to cheat, their body will naturally emit more heat due to increased mental and emotional effort. The rise in heat triggers the camera to focus on the participant's face and begin tracking their eye movements. The system then analyzes these data to assess the likelihood of cheating. While this study is fascinating, its implementation is notably complex [12]. Another study developed an exam cheating warning system using multiple-human pose estimation. The system works by recognizing the participant's head posture and hand movements that resemble cheating behavior. It is reported that the system successfully detects cheating behavior in both on-site (multi-user) and online (single-user) examination settings [13].

The implementation of deep learning methods in cheating detection systems for online exams has led to the development of more efficient and automated solutions. Object detection models based on CNN, such as YOLO v3 and v7, effectively identify unusual behavior of test takers. To determine whether this behavior is cheating, dual-channel CNN, convolutional neural networks (CNNs) and recurrent neural networks (RNNs) architecture, is employed for accurate classification [14], [15]. The CNN architecture has also been applied to detect cheating in paper-based examinations. A three-dimensional convolutional neural network (3D-CNN) is used to analyze CCTV video footage. This model effectively distinguishes between cheating acts, cheating attempts, and non-cheating actions in real-time [16].

2. Methods

An overview of the process for developing an intelligent model to detect fraudulent or suspicious activities during exams through surveillance camera video is shown in Fig 1. This study used Online Exam Proctoring (OEP) datasets from Michigan State University Computer Vision [9], which are publicly available online at <http://cvlab.cse.msu.edu/project-OEP.html> (accessed June 1, 2023). The process begins with video data preprocessing, then dividing the data into training and test sets using k-fold validation. Next, feature extraction is performed using various deep transfer learning techniques. The extracted features, with ImageNet initialization weights and retraining data, are then used to build a cheating detection model with several classification algorithms. Finally, the model is evaluated using test data, with performance metrics such as accuracy, recall, specificity, F1 score, and confusion matrix employed to assess its effectiveness.

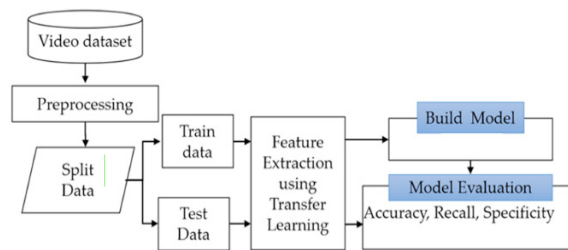


Fig. 1. Model development stage.

2.1. Dataset

Table 1. Data distribution of cheating exam video.

Label/ class to	Amount of Data
1	191
2	284
3	52
4	15
5	18

The OEP Dataset collection involved 24 participants, all of whom were students at Michigan State University. Among them, 15 were actors instructed to simulate exam-taking while performing cheating behaviors. These actors were not given specific directions on what forms of cheating to enact, allowing for naturalistic behavior. The remaining nine participants completed a real exam, with their scores being recorded. Since cheating was not possible in the monitored environment, the proctor prompted instances of cheating through means such as conversation, walking around, or handing out books. This combination of staged and real exam scenarios enriched the dataset with diverse cheating behaviors, enhancing its realism and variability. Each instance of cheating was annotated with three details: (a) start time, (b) end time, and (c) cheat type. Five categories of cheating were labelled: (1) use of books/notes/paper, (2) talking with someone in the room, (3) accessing the internet, (4) calling a friend, and (5) using a mobile device or other electronic devices. The dataset comprises a total of 560 video recordings, with the distribution of data across each cheating behavior category provided in Table 1.

2.2. Data pre-processing

To build a cheating exam detection model, the data must be preprocessed first. The video is decomposed into RGB frames at this step. The frames are then resized to the input dimensions required by the VGG model. After that, the pixel values of each frame set are extracted. As a result, the data is represented in a multidimensional matrix, where m indicates the number of videos and n indicates the number of frames extracted from each video. The output of this stage is a collection of images with dimensions of 224×224 pixels and three color channels (RGB), forming a tensor of $224 \times 224 \times 3 \times m \times n$, where m shows the number of videos, n is the number of images taken in each video.

2.3. Deep Transfer Learning Feature Extraction

a. Visual Geometry Group (VGG)

VGG-16 and VGG-19 are convolutional neural network (CNN) architectures developed by the Visual Geometry Group (VGG). These models are known for their simplicity and effectiveness in image classification tasks. Both architectures utilize small 3×3 convolutional filters, followed by max-pooling layers, which help extract hierarchical features from images. The primary difference between the two is the number of convolutional layers. VGG-16 consists of 16 layers, including 13 convolutional layers and three fully connected layers, whereas VGG-19 comprises 19 layers, consisting of 16 convolutional layers and three fully connected layers. As VGG-19 has more layers, it is capable of learning more complex features [17]. Both models employ the Rectified Linear Unit (ReLU) activation function in their hidden layers and are trained on extensive datasets such as ImageNet. This training allows fine-tuning across various tasks through transfer learning. While VGG-16 and VGG-19 are effective for image classification, they do have certain limitations. Their deep architectures result in high computational demands, requiring substantial memory and processing power. However, both models are less efficient at handling very deep networks because they lack advanced techniques such as residual connections [17].

b. Residual Network 50 Version 2 (ResNet-50 V2)

Residual Network 50 Version 2 (ResNet-50 V2) is one of the convolutional neural network (CNN) architectures that modify the original ResNet-50 architecture. ResNet itself stands for residual network because it introduces residual blocks, also known as skip connections, thus allowing the network to overcome the degradation problem that occurs in very deep networks more easily. Architectural improvements in version two have improved network performance and capabilities [18]. The architecture of ResNet-50 V2 is illustrated in Fig. 2.

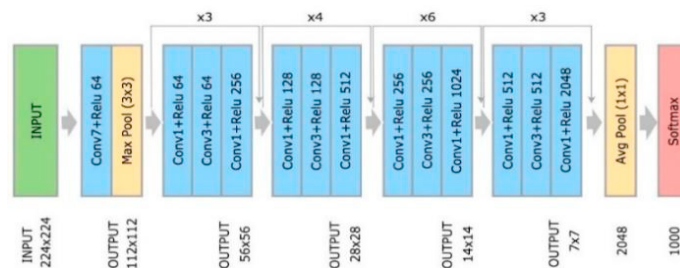


Fig. 2. Network architecture of Resnet-50 V2.

Residual blocks in ResNet-50 V2 enable the direct flow of information from the input layer to the next output layer, bypassing several intermediate layers. Hence, it helps alleviate the efficiency issue of training very deep networks. ResNet-50 V2 also uses bottleneck architecture on some of its residual blocks. The bottleneck architecture involves using a 1×1 convolution at the beginning to reduce the dimension of the feature, followed by a larger 3×3 convolution, and finally ending with another 1×1 convolution to restore the feature's dimension. ResNet-50 V2 implements batch normalization, which helps accelerate the training and stabilize the learning process by normalizing the layer output throughout the training. ResNet-50 V2 adopts a simpler residual block model compared to the "bottleneck" approach

used in the original ResNet. This simplification helps in reducing the number of parameters and computations required. ResNet-50 V2, which has been trained on large datasets, can be used as a transfer learning model for specific classification tasks on smaller datasets [8].

c. Inception

The Inception architecture was first introduced by Google researchers Szegedy et al. in 2017 [19]. This architecture represents a significant advancement in convolutional neural networks (CNNs) for image recognition. The core idea behind Inception is to enable the network to capture patterns and features at multiple scales within a single layer, which helps the model understand image details more efficiently. Unlike traditional CNNs, which use convolutional layers with a fixed kernel size in each layer that limits their ability to capture multi-scale patterns, Inception employs Inception modules in each layer. This approach enhances the network's flexibility, allowing it to perform multiple convolutions in parallel using different filter sizes, such as 1×1 , 3×3 , and 5×5 . Smaller filters capture fine details, while larger filters capture broader patterns, enabling the network to process different aspects of an image simultaneously. Each Inception module also includes a pooling operation to capture spatial hierarchies, making the network more robust to changes in scale and position within the image. To reduce the computational costs associated with these multiple convolutions, the architecture uses 1×1 convolutions as dimensionality reduction steps before applying larger convolutions. This process limits the number of channels or feature maps, preserving efficiency while maintaining high representational power. The outputs of all the different convolutions and pooling layers are then concatenated to form the output feature map of the Inception module, combining information from multiple scales in a way that enhances the model's flexibility and accuracy.

d. EfficientNetB7

EfficientNetB7 is a deep convolutional neural network model designed as part of the EfficientNet family, which scales the model dimensions (depth, width, and resolution) using a compound scaling method. This model architecture, introduced by Tan and Le [20], achieves higher performance with fewer parameters and lower computational costs compared to previous models. EfficientNetB7 represents the largest configuration in the EfficientNet series, offering improved accuracy while maintaining efficiency in both speed and memory usage. Its design and scalability make it particularly suited for applications requiring high-resolution image classification and transfer learning tasks in resource-constrained environments. EfficientNet models employ a compound scaling approach to optimize accuracy while maintaining low computational costs. This technique employs a structured formula to scale up all three dimensions simultaneously rather than scaling one dimension independently. EfficientNetB7 is built on a mobile inverted bottleneck convolution (MBConv) architecture combined with squeeze-and-excitation blocks. This architecture helps it to capture intricate spatial features with fewer parameters. Additionally, EfficientNetB7 utilizes batch normalization and Swish activation functions to enhance accuracy and convergence further.

2.4. Classification

a. Gated Recurrent Unit

Once feature extraction is completed, classification is performed using the Gated Recurrent Unit (GRU) algorithm. The GRU is a type of recurrent neural network (RNN) that builds upon the Long Short-Term Memory (LSTM) architecture but simplifies it by using a unit memory with only a reset gate and an update gate. This simplification is intended to make training and implementation easier while still preserving the LSTM's capacity to address the vanishing gradient problem. The reset gate determines whether the current state should incorporate information from the previous state, while the update gate controls how much information is carried over from the past state to the current state.

The first step in the GRU involves calculating the value of the update gate $u^{(t)}$ by substituting the input data $x^{(t)}$ and the previous hidden state $h^{(t-1)}$. The weight matrices and the bias vector are denoted as W_u , R_u , and b_u respectively. The mathematical formulation of the update gate is given by equation (1).

$$u^{(t)} = \sigma(W_u x^{(t)} + R_u h^{(t-1)} + b_u) \quad (1)$$

The following step determines the reset gate using the equation (2) and then substitutes the reset gate value to find the hidden layer candidate with equation (3), where W_r , is the weight and b_r is the bias.

$$r^{(t)} = \sigma(W_r x^{(t)} + R_u h^{(t-1)} + b_r) \quad (2)$$

$$\tilde{h}^{(t)} = \tanh(r^{(t)} \odot h^{(t-1)} R + x^{(t)} W + b) \quad (3)$$

The final step is to compute the output at time step t using equation (4). After developing the model, tests are carried out in real-time.

$$h^{(t)} = (1 - u^{(t)}) \odot \tilde{h}^{(t)} + u^{(t)} \odot h^{(t-1)} \quad (4)$$

b. Long Short-Term Memory (LSTM)

Long Short-Term Memory (LSTM) networks are a type of recurrent neural network (RNN) architecture specifically designed to effectively capture long-term dependencies in sequential data, including text, audio, and time series data. Traditional RNNs often encounter difficulties with vanishing or exploding gradients during the training of long sequences, which limits their capacity to retain information over time. LSTMs overcome this issue by utilizing a cell state that preserves information across time steps, along with a series of gates (a forget gate, an input gate, and an output gate) that regulate the flow of information into, out of, and within the cell. The forget gate controls which information is stored and discarded, computed by equation (5). The input gate updates the memory cell contents using equation (6).

$$f_t = \sigma(x_t U^f + h_{t-1} W^f) \quad (5)$$

where x_t is the input data at the time step t , h_{t-1} the cell status of the previous time step, and U^f , W^f are the weights.

$$i_t = \sigma(x_t U^i + h_{t-1} W^i) \quad (6)$$

where U^i and W^i are weights. The output gate can be found by equation (7).

$$o_t = \sigma(x_t U^o + h_{t-1} W^o) \quad (7)$$

The next step is to define new information $\hat{C}_t$ with equation (8) and update C_t with equation (9).

$$\hat{C}_t = \tanh(x_t U^g + h_{t-1} W^g) \quad (8)$$

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \hat{C}_t \quad (9)$$

Then C_t is used to compute h_t , the hidden state at time step t , using equation (10).

$$h_t = \tanh(C_t) \cdot o_t \quad (10)$$

c. Convolutional Neural Network (CNN)

A Convolutional Neural Network (CNN) is one of the most widely used deep learning architectures in visual data processing, such as images and videos. CNN is designed to automatically extract and learn feature representations from input data through a series of layers, including convolution, non-linear activation, pooling and classification operations. The key component of CNN is the convolution layer, which consists of multiple small kernels (or filters) (e.g., 3×3 or 5×5) to perform convolution operations on the input. The two-dimensional convolution operation between the input X and the kernel K expressed by the equation (11),

$$S(i, j) = (X * K)(i, j) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} X(i+m, j+n) \cdot K(m, n) \quad (11)$$

where M and N are the filter sizes, X is the input image matrix, and $S(i, j)$ is the output at location (i, j) . These filters are designed to extract spatial features from images, such as edges, textures, and patterns. After that, a non-linear

activation function, such as Rectified Linear Unit (ReLU), is applied element by element. This introduction of non-linearity allows the network to learn complex representations and helps mitigate the vanishing gradient problem. The next layer in most CNN models is the pooling layer, which reduces the spatial dimension of the feature map. This dimension reduction aims to improve computational efficiency and make feature representations more robust against small translations in input data. The most widely used pooling technique is max pooling, which selects the maximum value from a specified local window. Once the spatial features are compacted, the results are flattened and forwarded to one or more fully connected (FC) layers. These layers integrate all the learned features to generate the final output, such as class probabilities in a classification task. The SoftMax activation function is commonly used for multi-class classification tasks. The CNN model is then trained using the backpropagation algorithm to minimize the loss function. During the training process, the weights within the convolutional kernels and fully connected layers are fully updated using optimization algorithms, such as Stochastic Gradient Descent (SGD).

3. Result and Discussion

3.1. Experiment Results

The performance of the proposed algorithm, which combines deep transfer learning with a classification model, was compared using four metrics derived from the confusion matrix. The four metrics are accuracy, precision, recall, and F1 score. Accuracy refers to the ratio of correctly predicted instances to the total number of instances. Sensitivity or recall indicates the proportion of correctly identified positive cases among all actual positive instances. Precision is the proportion of correctly identified positive cases to all instances that were predicted as positive. The F1 score is the harmonic mean of precision and recall.

Table 2. Performance of cheating exam detection model.

Method	Feature Extraction	Accuracy	Recall	Precision	F1-Score
LSTM	VGG16	0.5446	0.2000	0.1089	0.1410
	VGG19	0.7321	0.3468	0.2893	0.3132
	Resnet50V2	1.0000	1.0000	1.0000	1.0000
	InceptionResNetV2	0.9643	0.8913	0.9070	0.8853
	EfficientNetB7	0.5446	0.2000	0.1089	0.1410
GRU	VGG16	0.7679	0.3550	0.3011	0.3255
	VGG19	0.7054	0.3296	0.2776	0.3007
	Resnet50V2	1.0000	1.0000	1.0000	1.0000
	InceptionResNetV2	0.9464	0.9630	0.9255	0.9398
	EfficientNetB7	0.5446	0.2000	0.1089	0.1410
CNN	VGG16	0.5446	0.2000	0.1089	0.1410
	VGG19	0.5446	0.2000	0.1089	0.1410
	Resnet50V2	0.7232	0.4159	0.4178	0.4151
	InceptionResNetV2	0.6518	0.2734	0.2811	0.2579
	EfficientNetB7	0.5446	0.2000	0.1089	0.1410

The evaluation of all models' performance by comparing these four metrics is conducted with careful consideration of the data quantity among the five classes of cheating actions. In the case of data imbalance, accuracy alone is not enough to guarantee the overall model performance. Consider Table 2, which compares F-1 scores and the accuracy of various algorithms. The VGG16-LSTM algorithm achieved an accuracy of 0.5446; however, its F1-score was considerably lower at 0.141. A similar discrepancy between accuracy and F1-score is evident in other models, such as VGG16-CNN and VGG16-GRU, highlighting a significant gap. This suggests that some minority classes are being misclassified. A more detailed analysis can be seen in Table 2, which shows that recall and precision are consistent with the reported F1-score values.

As shown in Table 2, the LSTM-Resnet50V2 and GRU-Resnet50V2 models achieved perfect scores of 1.0 for F-measure, recall, and precision. In contrast, the CNN model demonstrated the lowest performance and failed to address the class imbalance present in the dataset. Compared to other studies that used the same dataset, the results achieved

in this study were superior [18]. In terms of model training time and testing time, as shown in Figures 3 and 4 respectively, the ResNet50V2-GRU model demonstrates faster performance compared to the ResNet50V2-LSTM.

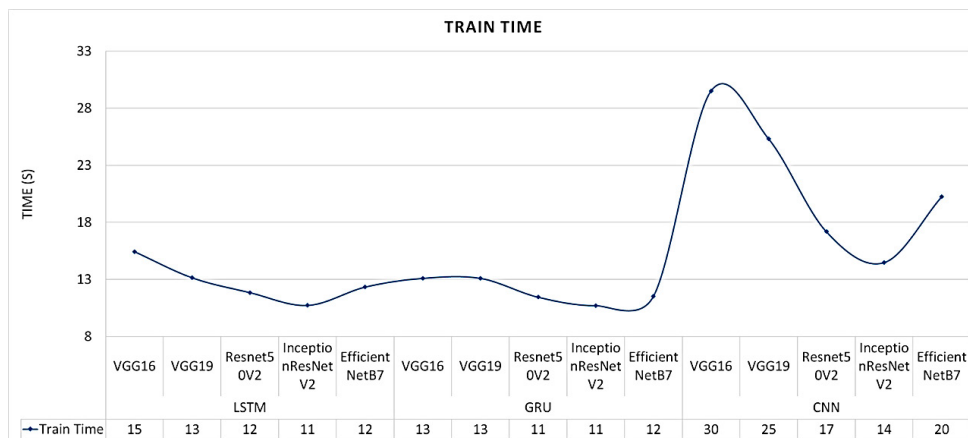


Fig. 3. CPU Time of the Training Phase of Each Algorithm

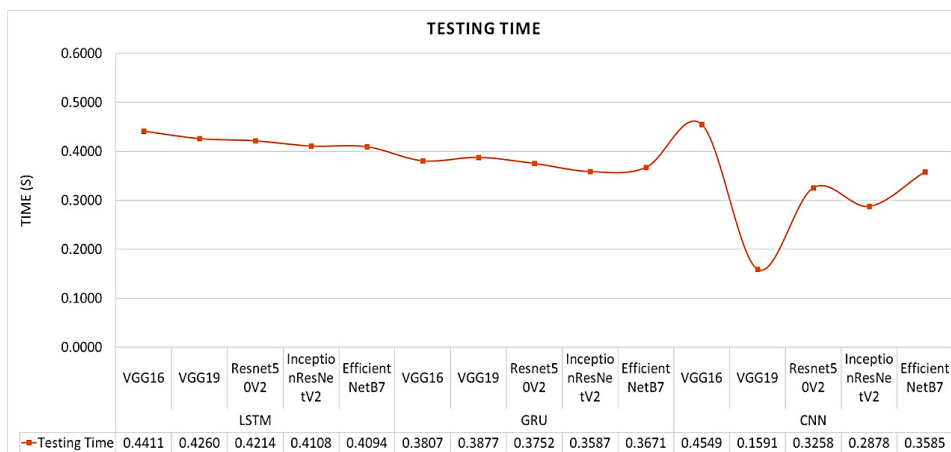


Fig. 4. CPU Time of the Testing Phase of Each Algorithm

Table 3. Comparison Performance of Cheating Exam Detection Model with Previous Similar Studies

Reference	Method	Dataset	Accuracy
Atoum, Y. et.al [21]	PCA + SVM	MSU OEP Dataset	0.942
Imal, E. et.al [22]	ResNet50 + LSTM	MSU OEP Dataset	0.964
S. Kaddoura, et.al [23]	DFT + CNN	MSU OEP Dataset	0.983
Puspitasari, RDI, et.al [24]	Resnet50 + CNN	MSU OEP Dataset	1.000
Proposed Method	Resnet50 + GRU	MSU OEP Dataset	1.000

Table 3 presents a comparative performance analysis of various methods for cheating detection in exam scenarios using the MSU OEP Dataset. All the methods utilize different deep learning or machine learning architectures. Earlier approaches, such as the PCA and SVM models by Atoum et al. [21], achieved an accuracy of 0.942, indicating the effectiveness of classical machine learning methods. More recent studies have adopted deep learning approaches. Imal et al. [22] used ResNet50 and LSTM, achieving an accuracy of 0.964, showing the potential to combine CNN-based feature extraction with temporal modeling through LSTM. S. Kaddoura et al. [23] improved accuracy further with combining frequency domain information (DFT) and convolutional learning (CNN) model, achieving an accuracy of

0.983. The study by Puspitasari et al. [24] and the proposed method in the current research report perfect accuracy of 1.000 using ResNet50 and CNN and ResNet50 and GRU, respectively. Both results suggest that ResNet50 is a highly robust feature extractor for this dataset, while both CNN and GRU can capture relevant temporal and spatial patterns. A perfect 1.000 accuracy rate indicates that the model has learned the dataset too well. Nevertheless, future work should validate robustness and generalizability in real-world settings to ensure the model's reliability.

4. Conclusion

The study examined various deep transfer learning feature extraction methods and classification algorithms to automatically classify five cheating behavior classes from the Cheating Exam Detection dataset, provided by Michigan State University's Computer Vision Lab. The dataset consists of 560 videos with an imbalanced data distribution across five cheating behavior categories: opening a book, talking to someone, accessing the internet, making a phone call, and using a phone or other electronic device. Experiments were conducted by exploring the features of VGG-16, VGG-19, Resnet-50 v2, and Inception-Resnet v2 when classified using LSTM, GRU, and CNN. The experimental results reveal that the ResNet50V2 setup yields the best overall performance, with the GRU and LSTM variants attaining the highest F1 scores and accuracies. The examination of model training and testing duration indicates that ResNet50V2-GRU model performs faster than ResNet50V2-LSTM model. Overall, the ResNet50V2-GRU model exceed all the tested model in the paper in term of the classification effectiveness, model training time, and model testing time. Among the tested feature extraction models, ResNet-50V2 yielded the best results. The study found that deep transfer learning can effectively handle imbalanced datasets without requiring data resampling during preprocessing

Author Contribution Statement

The authors' contributions are as follows: E. M. Imah contributed to conceptualization, investigation, original draft preparation, supervision, and data analysis. R. D. I. Puspitasari developed the program, conducted the experiment, and wrote the data comparison results. F.Q. Annisa wrote the methods in the paper, revised some parts of the paper, and performed data preprocessing. H. Prayogi contributed to analyzing the data, revised the paper, and conducted a literature study with I. Fitriyaningsih. H. Prayogi and I. Fitriyaningsih also contributed proofreading and editing of the layout.

AI Usage Declaration

In preparing this manuscript, we utilized ChatGPT to assist with translation and refine phrasing in some parts of the Methods section only. We declare that we use the AI responsibly and heavily supervise the results from the AI tool.

Data Availability

Data utilized in this research is available to the public through the following URL: <https://cvlab.cse.msu.edu/oep-dataset.html>

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